Rishitha Naga Durga Gona

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Summary

Computer Science graduate (B.Tech, Artificial Intelligence and Machine Learning, SRM University AP, 2026) with a strong foundation in computer science fundamentals and practical experience across machine learning, natural language processing and LLM-based applications. Skilled in Python-based data analysis, model training and evaluation, and building interactive applications that turn data into usable insights. Quick to learn new technologies and committed to solving real-world problems with well-evaluated solutions.

Education

B.Tech - Computer Science Engineering (AIML), SRM University AP 2026 | GPA: 7.3/10

Intermediate - MPC, Bhashyam Junior College 2022 | 86.5%

Secondary Education — Bhashyam Public School 2020 | 98%

Work Experience

Full Stack Java Intern | Edubot Technologies June–July 2024

Technologies: Java, Servlets, JSP, PostgreSQL, SQL, Eclipse

- Developed a Java-based Electricity Billing System to automate customer, billing and usage management.
- Built Java Servlets and JSP pages to handle requests, process data and render dynamic web interfaces.
- Integrated PostgreSQL for customer and billing data, implementing CRUD operations through SQL queries.
- Implemented registration, login and data-entry forms with input validation and error handling.
- Performed functional testing and debugging to identify issues and ensure the system worked reliably.

Skills

AI & Machine Learning: Machine Learning, Deep Learning, Neural Networks, Natural Language Processing (NLP), Transformers, Model Training & Evaluation, Feature Engineering, Data Preprocessing

Generative AI & LLMs: Generative AI, Large Language Models (LLMs), AI Agents (Tool Calling), Agentic AI, Prompt Engineering

Frameworks & Libraries: PyTorch, Hugging Face Transformers, Scikit-learn, LightGBM, Pandas, FastAPI, Streamlit, Gradio

Programming Languages: Python, Java, C, C++, JavaScript, SQL

Web & Databases: HTML5, CSS3, React, PHP, PostgreSQL, SQLite

Tools: Git, GitHub, VS Code, Eclipse, Claude Code, OpenAI Codex

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Debugging and Testing

Projects

Data Insight AI: LLM Data Analyst AI Project

Technologies: Python, Gemini LLM, Tool Calling, Pydantic, Streamlit, Pandas, Scikit-learn, Statsmodels, Plotly

- Developed an LLM-powered data analysis application that converts any CSV file into an interactive dashboard with KPI cards and cross-filtering charts, 3-month forecasts and an AI-generated business brief.
- Implemented an agentic workflow using LLM tool calling to run sandboxed Pandas computations for business questions, ensuring every reported figure is computed rather than generated.
- Prevented hallucinated figures by enforcing structured LLM outputs (JSON schema with Pydantic validation) and grounding prompts in pre-computed signals only.
- Engineered a multi-model LLM gateway with automatic fallback on rate limits, sustaining AI features beyond Gemini's 20-requests/day free-tier cap.
- Built a resilient CSV ingestion pipeline (mixed encodings, delimiters, currency/percent text; 300K rows in under 1 second) and time-series forecasting with uncertainty ranges, fixing a defect that produced a −27% forecast for growing revenue.

Ghar AI: House Price Prediction AI Project

Technologies: Python, LightGBM, SHAP, Feature Engineering, FastAPI, SQLite

- Developed an end-to-end machine learning platform that predicts fair prices for 23,000+ real estate listings across 250 cities in India and flags good deals and overpriced homes.
- Trained a LightGBM gradient boosting model that achieved 12.7% median error, outperforming a broker-style baseline and linear regression, with 5-fold cross-validation to prevent data leakage.
- Engineered a K-nearest-neighbour comparable-homes feature, the model's strongest predictor, and cleaned 29,451 raw listings, improving R^2 from 0.40 to 0.89.
- Integrated explainable AI (SHAP) to justify every predicted price, and quantile regression to produce price ranges containing the true price 81% of the time.
- Deployed the model through a FastAPI backend with interactive map search, an EMI and total-cost calculator (stamp duty, GST, RBI loan-to-value rules) and a validated enquiry system on SQLite.

Hybrid Humor Detection in Code-Mixed Text Capstone Project

Technologies: Python, PyTorch, Hugging Face, MuRIL, XLM-RoBERTa, IndicBERT, Scikit-learn, Gradio

- Created a balanced 10,000-sample Tenglish (Telugu-English) humor dataset with 6 humor categories and Romanized spelling variants, addressing the lack of public code-mixed humor data.
- Fine-tuned and benchmarked 10 transformer models (MuRIL, XLM-RoBERTa, mBERT, IndicBERT, DeBERTa and others) using stratified 80/10/10 splits and F1-based model selection.
- Designed a hybrid architecture combining fine-tuned MuRIL with a rule-based heuristic engine (emojis, English and Telugu slang, stylistic cues), achieving 92.4% accuracy and 0.918 F1.
- Demonstrated that Indian-language pre-trained models outperform English-only models on code-mixed text (MuRIL 90.1% vs BERT 74.5%), with the heuristic layer adding a further 2.3 points.
- Deployed the final model as a Gradio web application with real-time predictions, confidence scores and explanations of the triggered humor cues.

Implicit Customer Sentiment Analysis Research Project

Technologies: Python, PyTorch, Hugging Face, RoBERTa, BiLSTM, Scikit-learn, Gradio

- Designed a hybrid deep learning model (RoBERTa embeddings, BiLSTM and attention) to detect implicit sentiment in movie reviews, based on published research (Applied Soft Computing, 2024).
- Achieved 89.6% accuracy on the IMDb 50K dataset, outperforming a TF-IDF and Logistic Regression baseline (88.6%).
- Optimized GPU training with a frozen RoBERTa encoder, FP16 mixed precision, AdamW with learning-rate warm-up and checkpointing.
- Performed exploratory data analysis on 50,000 reviews (class balance, review length, word frequency) and built the text preprocessing and tokenization pipeline.
- Deployed the trained model as a Gradio web application for real-time sentiment prediction.

Certifications

AI Fundamentals - Coursera	2026
AWS Certified Cloud Practitioner	2025
NPTEL – The Joy of Computing Using Python (IIT Madras)	2023